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To:
Guest Editors, Special Collection on "Ethics of Digital Transformation in Tourism and Hospitality"
Editors-in-Chief, *Information Technology & Tourism*, Springer Nature

Subject: Submission of Original Research Article for Special Collection Consideration

Dear Guest Editors and Editorial Board,

I am pleased to submit my original research manuscript entitled **"Explainable Hotel Booking Cancellation Prediction Models: Towards Trustworthy Survival Analysis in Hospitality"** for consideration in *Information Technology & Tourism* within the Special Collection on **"Ethics of Digital Transformation in Tourism and Hospitality"**.

As digital platforms and intelligent intermediaries increasingly govern hospitality operations, delegating capacity allocation and pricing to opaque "black-box" predictive models creates severe ethical vulnerabilities. Miscalibrated attrition forecasts lead directly to aggressive overbooking, guest displacement, breach of consumer trust, and acute consumer insecurity. Traditional hospitality literature frames cancellation forecasting as a static binary classification problem, which ignores the temporal progression of cancellation risks and introduces systematic predictive bias by treating active, right-censored reservations as non-canceled.

To operationalize the core principles of ethical artificial intelligence, transparency, and algorithmic accountability in hospitality, this study introduces an explainable time-to-event (survival analysis) framework delivering the following core contributions:

1. **Continuous Survival Formulation without Imputation Bias:** We reframe cancellation forecasting as a continuous time-to-event problem evaluated on a large-scale empirical cohort with verified timestamps ($N = 119{,}390$), completely eliminating synthetic imputation artifacts.
2. **Probability Calibration as an Ethical Safeguard:** Through 100-replicate non-parametric bootstrap resampling, omnibus Friedman testing, and empirical reliability diagrams across operational planning horizons ($30, 60, 90$, and 180 days), we uncover a critical trade-off: while standard Random Survival Forests maximize discriminative ranking, Oblique Random Survival Forests achieve superior probability calibration. Calibrated probabilities provide hospitality managers with an ethics-by-design mechanism that prevents erratic overbooking and mitigates consumer insecurity.
3. **Continuous XAI for Algorithmic Auditing:** We introduce time-dependent Survival Permutation Feature Importance and Partial Dependence Survival Profiles. These continuous explainers decode non-linear risk dynamics and successfully detect structural policy anomalies, unmasking why non-refundable rates exhibited elevated cancellation hazards due to speculative intermediary block bookings.

Special Collection Guest Editors:

- **Dr. Boyu Lin** (Guest Editor), Macao University of Tourism, Macao.
- **Prof. Woojin Lee**, Arizona State University, USA | **Prof. Nao Li**, Beijing Technology and Business University, China | **Prof. Lu Lu**, Temple University, USA.

Suggested Potential Reviewers:

- **Nuno Antonio, Ph.D.** – Nova Information Management School (NOVA IMS), Lisbon, Portugal.
  Email: nantonio@novaims.unl.pt | *Expertise:* Hotel cancellation prediction, machine learning in hospitality analytics.
- **Martin Falk, Ph.D.** – School of Business, University of South-Eastern Norway, Norway.
  Email: martin.falk@usn.no | *Expertise:* Tourism economics, booking channel dynamics, econometric cancellation modeling.
- **Apostolos Ampountolas, Ph.D.** – Boston University, School of Hospitality Administration, USA.
  Email: aampount@bu.edu | *Expertise:* Capacity governance, hospitality demand forecasting, predictive analytics.

Author Declarations:

- This manuscript represents original work not published previously and not currently under consideration elsewhere.
- The author approves the manuscript and agrees with its submission to *Information Technology & Tourism*.
- The author declares no conflicts of interest or competing financial interests with the suggested reviewers.
- To ensure open science, auditability, and full reproducibility, all source code and experimental pipelines are publicly available at: GitHub Repository (mcavs/explainable-hotel-booking-cancellation-predictive-models).

Thank you very much for your time, consideration, and editorial handling of this manuscript.

Kind regards,

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